

Experiment : 7

TO FIND ROOTS OF QUADRATIC EQN

AIM

To write a program to find the roots of a quadratic equation

ALGORITHM

1) Read $a, b, c, d, x_1, x_2, x_1 \text{img}, x_2 \text{img}$

2) $d = (b*b) - (4*a*c)$

3) If $(d == 0)$

$$x_1 = -b / (2*a)$$

$$x_2 = x_1$$

else if $(d > 0)$

$$x_1 = (-b + \text{sqrt}(d)) / 2a$$

$$x_2 = (-b - \text{sqrt}(d)) / 2a$$

$$x_1 \text{img} = 0$$

$$x_2 \text{img} = 0$$

else

$$d = -d$$

$$x_1 = -b / 2*a$$

$$x_1 \text{img} = (\text{sqrt}(d)) / 2a$$

$$x_2 = x_1$$

$$x_2 \text{img} = (\text{sqrt}(d)) / 2a$$

4. Print roots as $\%d + \%di$ and $\%d - \%di$

OUTPUT

Enter the values of a, b, c

$$(2 \times 2) - (4 \times 1 \times 1) = 0$$

d=0

The roots of quadratic eqn is -1.00000 and -1.00000

ALGORITHM

1. Read a, b, c
2. $d = (b \times b) - (4 \times a \times c)$
3. If $d < 0$ then
4. $x_1 = (-b + \sqrt{d}) / (2 \times a)$
5. $x_2 = (-b - \sqrt{d}) / (2 \times a)$
6. Print x1, x2
7. Else if $d = 0$ then
8. $x = -b / (2 \times a)$
9. Print x
10. Else $d > 0$ then
11. $x_1 = (-b + \sqrt{d}) / (2 \times a)$
12. $x_2 = (-b - \sqrt{d}) / (2 \times a)$
13. Print x1, x2
14. End

RESULT

A C program to find the roots of a quadratic eqn was compiled.